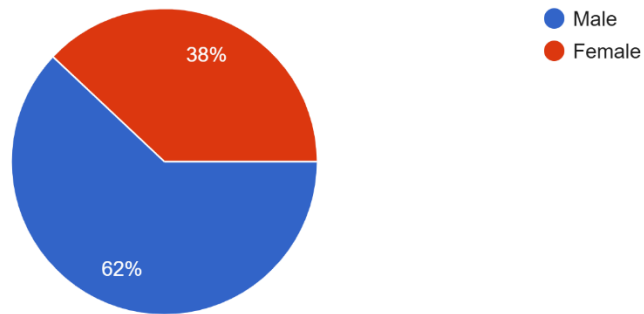


FINGRATOR RESPONSES GROUP 5 Survey results

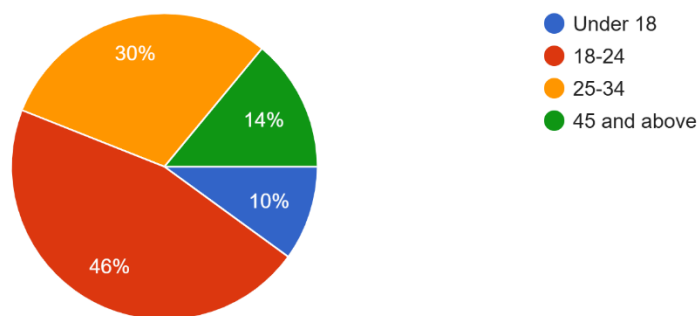
Gender

50 responses



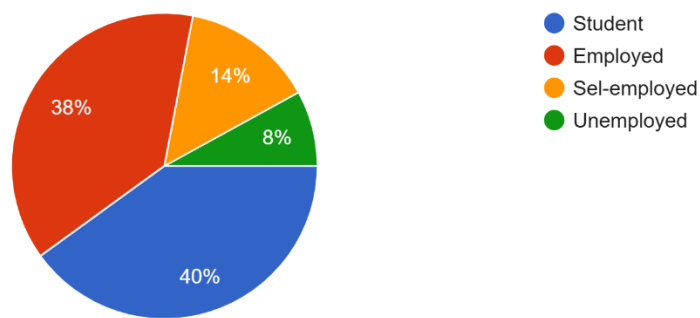
1. What is your age group?

50 responses



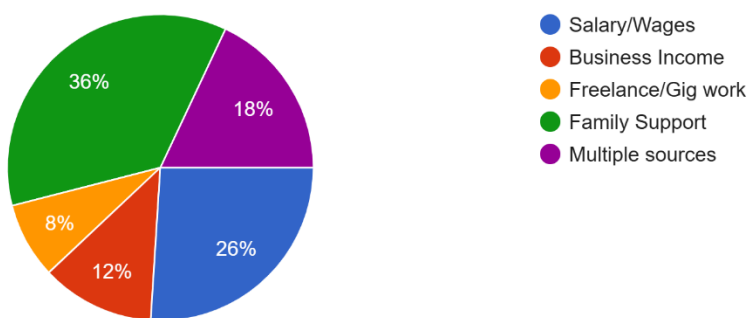
2. What is your employment status?

50 responses



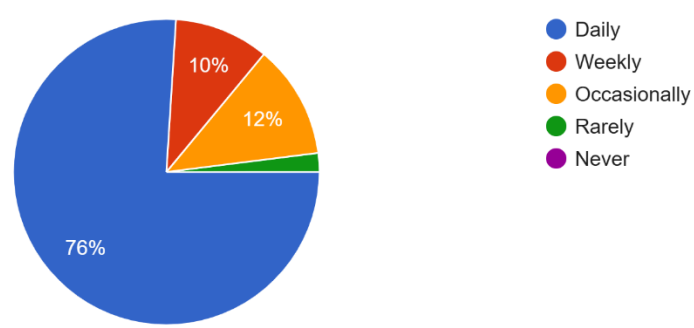
3. Which of the following best describes your primary source of income?

50 responses



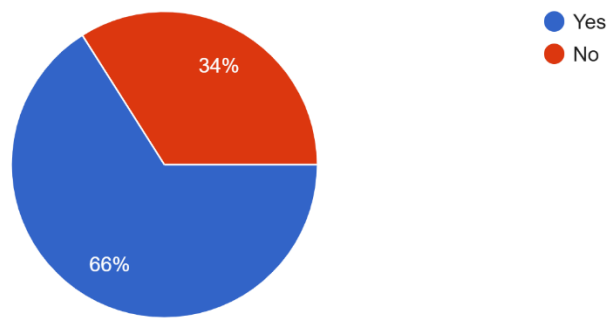
4. How often do you use mobile money services (eg; MTM Momo, Telecash, AirtelTigo Money)?

50 responses



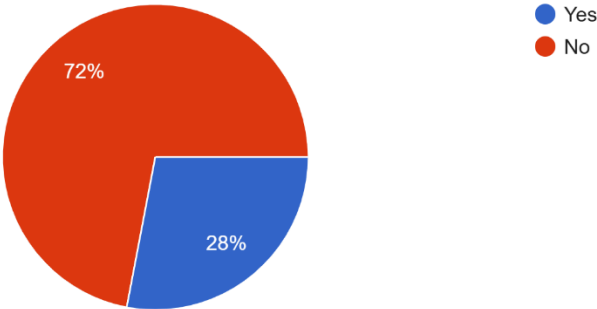
5. Do you actively use a bank account?

50 responses



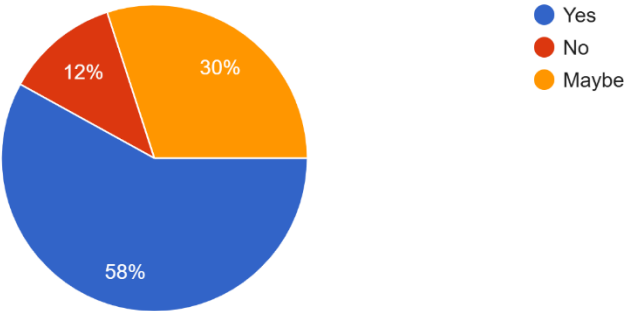
6. Have you ever used a financial or budgeting app before?

50 responses



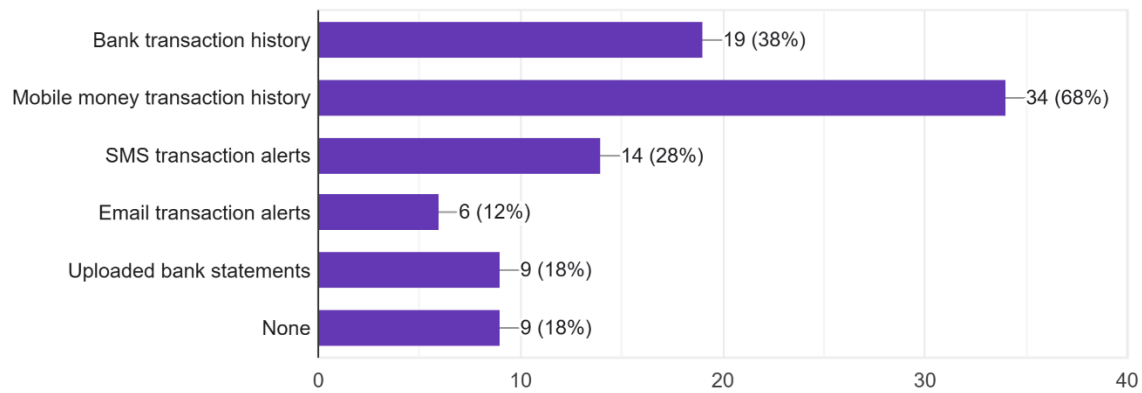
7. Would you allow a financial app to analyze your transactions in order to provide financial insights and recommendations?

50 responses



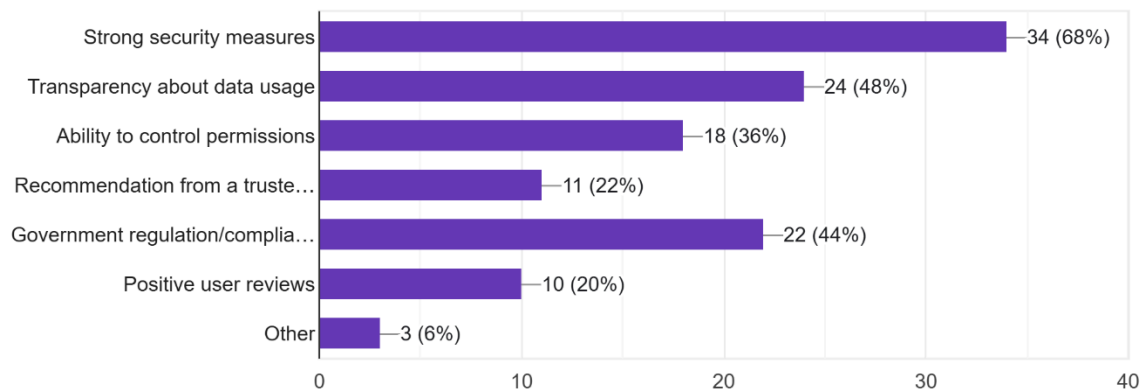
8. Which financial data would you be willing to share with such an app? (Select all that apply)

50 responses



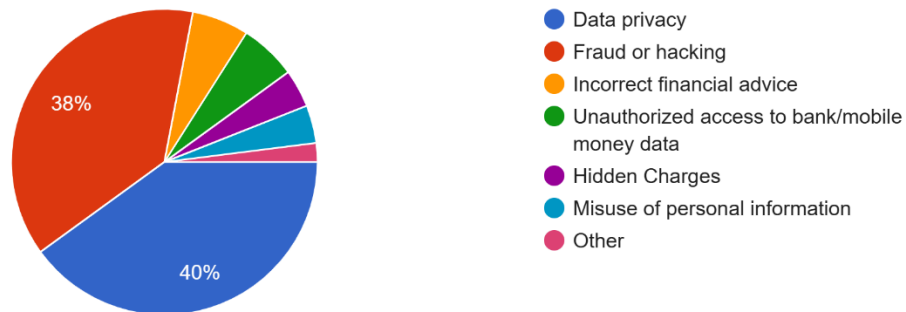
9. What would make you feel comfortable sharing your financial data with an app? (Select up to 2)

50 responses



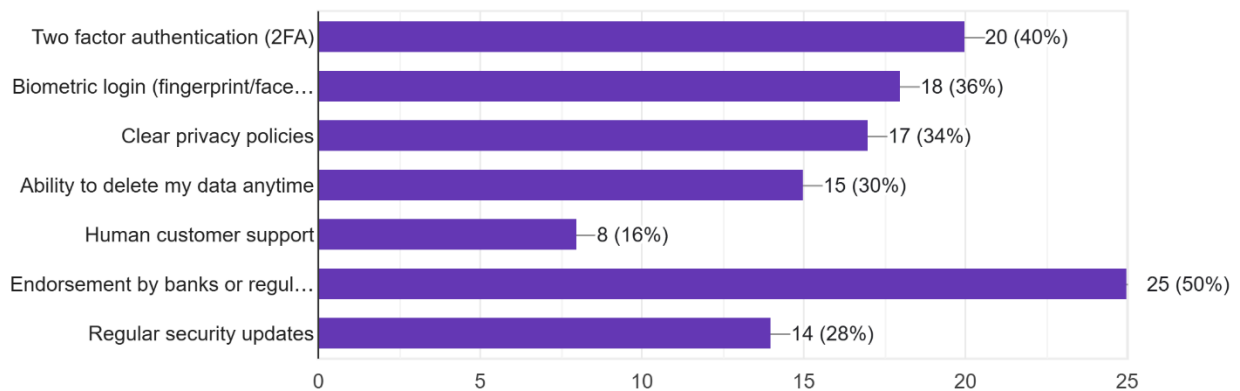
10. What concerns you most about using a financial intelligence app?

50 responses



11. Which of the following would increase your trust in a financial app? (Select up to 2)

50 responses



12. What should a financial app **NEVER** do with your data?

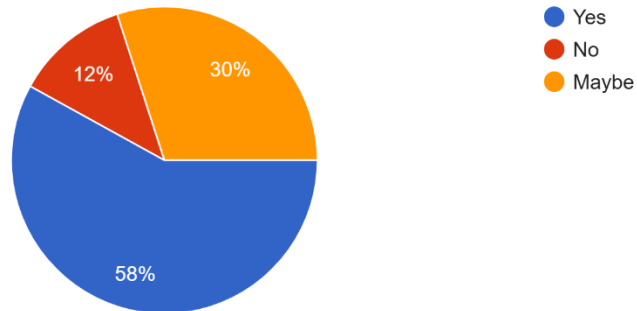
Share
Share it with third parties
Use my data for commercial purposes
Never to share my data with the third party
A financial app should never sell my personal data, share my banking credentials, store sensitive information insecurely, or use my financial activity for advertising without my clear consent
Share with other people
Give it out without authorization
Leak

Unauthorized use of my data
Share or sell it
Misuse transaction history
sell it
Share it
Share it with other people
Profit from my data
Share my data without my permission
Sell it
Leak confidential financial records
Not to share my data with unauthorized person
Publish without my consent
Use it for their own benefits
Make it public
Unlawfully sell it to an unknown party
Store without my permission
Use data for unauthorized marketing
Sharing without permission
Never share or sell my data
Distribute
When it takes other information I didn't sign up for
Expose my identity online
Well I will honestly say you shouldn't ignore the PCIDSS framework. We know you could be fined at maximum 18000 but if you would want to go international and you ignore the framework you could be fined as much as \$5000 to \$100000. Although you may allow user to delete their data, keep a copy maybe not the full thing but a small copy just in case your user turns back to frame you for something. So my advice is keep it safe ensure no data is enclosed or made available to an unauthorized individual (just saying follow the CIA trail as well). Wish you luck on your academic research
Share
share with a third party without informing me
No
Use the data for their own personal gain or share with a third party
Violate data protection laws
It must allow free flow of the app but not the use of data
Save and share without my permission
Share it to a third party
Access my account without permission
Leak it
Distribute illegally
spy on it
Generate false data
Use it to make payments I'm not aware
Track my activity secretly

Share personal info
Never share my data

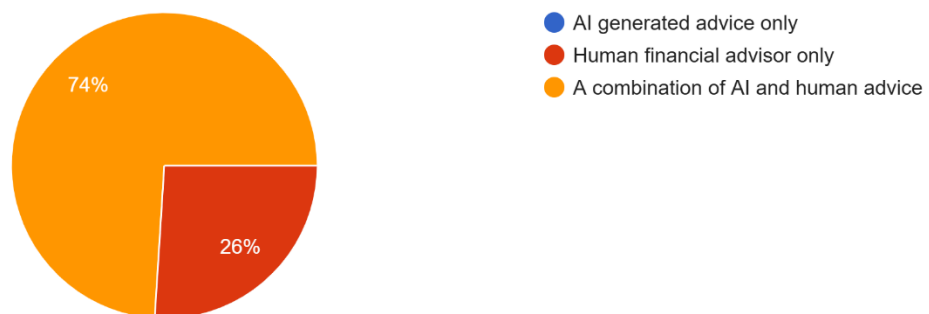
13. Would you use an AI-powered financial assistant that gives spending insights, savings advice, and cash flow forecasts?

50 responses



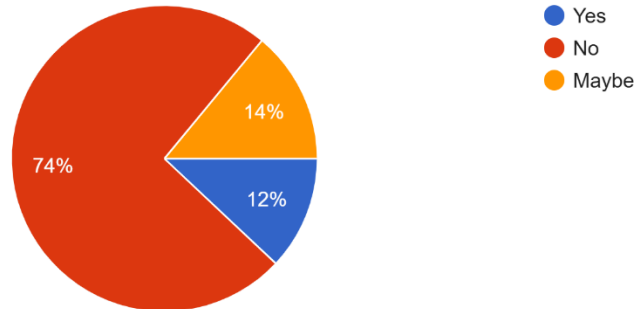
14. Which type of financial advice would you trust more?

50 responses



15. If the app occasionally makes inaccurate predictions or recommendations, would you continue using it?

50 responses



16. In your opinion, what is the most important feature a financial intelligence app should have?

Security
Data privacy
Security
Automated, real-time transaction tracking and categorization combined with predictive AI insights
Multi-layered protection and data privacy.
Strong security
Safety
Flexible to use
The app should provide clear, personalized insights that helps the user to make smarter financial decisions.
Proper security systems
Constant updates
Predictive analytics
it should be secure
Two factor authentication
Reliable calculation
Constant information updates
Thorough terms and conditions(policies)
24/7 accessibility and support
Absolute results
Higher accuracy and precision in it's ability to predict.
Biometric features
Provide accurate information
Transparency

Secure cloud backup
Strong analysis skills
The ability to give precise info and advice in every change in a situation or variable
Data security
Analysis of spending and how to save
Personalized recommendations
To be updated on all financial interest rates for people who want to invest, ability to maybe not fully but understand the market trend for stocks investors. And mostly understand the needs or what the user mostly spends their money on that is if it's either relevant or irrelevant, either an asset or just a waste of time.
How to multiply my money
ability to manage and safeguard my data
I don't know
Reliable
Accurate budgeting tools
Allow to learn more about things we do not understand
The app should have constant human review
It should be able to give good predictions
Good security
Simple and user-friendly design
Policy transparency
Tight security
i don't know
Give me real time information
Fast transaction tracking
Accurate information
Financial analysis and forecasting



UNIVERSITY OF GHANA

UNIVERSITY OF GHANA BUSINESS SCHOOL: DEPARTMENT OF ACCOUNTING
MSAF 626; DIGITAL FINANCE
FINGRATOR PROJECT

GROUP 5 REPORT

GROUP MEMBERS:

SAMSON KOJO OBRO - 22417106

MONICA AVARIPO - 22259829

ERIC OSAE ODURO JUNIOR – 22418672

EUNICE AMPONSAH - 22418691

1.0 INTRODUCTION

Digital finance developments have made changes in the way people manage their money and financial services in the global setting. In Ghana, development in mobile money services, digitized banking, and innovations in fintech have brought changes to the financial environment. From information provided by the Bank of Ghana, it is evident that there has been a continued growth in transactions carried out through mobile money services, which indicates reliance on digital financial services for individuals and business organizations (Bank of Ghana, 2024). Nonetheless, users are yet to develop skills for effective budgeting, saving, financial projections, and financial planning in general.

Conventional budgeting applications are mostly limited in terms of monitoring expenses incurred in the past but fail to assist users with financial decisions. FinGrator Pulse aims at making changes by acting as an intelligent financial guidance application and not an ordinary budgeting application. The software monitors data from various sources including banks' accounts, mobile money accounts, text messages, email messages, and uploaded files to predict future finances.

This report highlights the importance of financial logic, data management, and trust-building processes vital to successful implementation of FinGrator Pulse. The research employs survey questionnaires administered to 50 respondents to analyze attitudes of users towards data sharing, AI-powered finance advisory services, issues related to privacy protection, predictive analytics, and general trust in digital finance systems.

Under the premise of the Technology Acceptance Model introduced by Davis in 1989, it can be argued that the more useful and user-friendly technology is considered by potential users to be,

the more willing they are to adopt it. Moreover, according to the Theory of Planned Behavior introduced by Ajzen in 1991, the combination of users' attitudes, levels of their trust, and perceived behavioral control affects the likelihood of adoption of digital finance technology.

The report is guided by recent fintech adoption studies which suggest that users are more likely to adopt digital financial technologies when they perceive them as useful, trustworthy, secure, and easy to use (Al-Okaily et al., 2022). The report also draws on contemporary digital trust and behavioral finance research which argues that user attitudes, perceived control, security assurance, and trust significantly influence adoption of AI-driven financial systems (Kaur et al., 2021).

Unlike theoretical research studies, the current study takes a practical fintech product approach. Thus, the research implications are developed in line with the goal to create a functional financial intelligence platform for practical use.

2.0 ANALYSIS OF THE DIGITAL FINACE ENVIRONMENT

Financial behaviors have been redefined by the advancement of fintech and digital financial services. For instance, in Ghana, mobile money services have been identified as some of the leading technologies because of their convenient nature and capacity to facilitate financial inclusion. Financial inclusion can be defined as availability and utilization of cheap financial services offered to individuals and firms (Demirgüç-Kunt et al., 2022).

It is clear from the findings of the survey that people use financial services in digital form. Seventy-six percent of participants confirmed their use of mobile money services every day. This shows that they are well aware of the importance of utilizing their financial resources in a digitized manner. This finding supports findings by Evans & Pirchio (2015). According to these authors, mobile money solutions thrive in developing countries because of their success in lowering the costs of transactions and financial access.

However, financial management has remained disjointed for most users. Survey results showed that 36% of respondents depended on freelance or gig work while 18% received income from various sources. This shows unstable income flows from various sources for many people. This problem arises because people have a tendency to make poor and sometimes inconsistent decisions due to uncertainty about the future according to Kahneman and Tversky's Prospect Theory (1979).

While most financial management apps rely on past transactional history, FinGrator Pulse attempts to address the gap through the introduction of predictive analytics into personal financial management applications. Predictive analytics can be described as a combination of historical data, statistical algorithms, and artificial intelligence that can be used for forecasting future results (Shmueli & Koppius, 2011).

In turn, the results demonstrate that the trust problem persists as a major issue with fintech adoption. The users' data is extremely personal and private, and many people tend to be reluctant to reveal their transactional history or financial information. While a large percentage of the surveyed individuals stated that they would let an app analyze their transaction, 30% of the responses were "Maybe." This is probably due to their concerns about trust and privacy.

This finding supports recent fintech trust studies which show that trust significantly influences users’ willingness to engage with digital systems involving financial transactions and AI-based financial recommendations (Chandra et al., 2022). Users are more likely to adopt fintech platforms when they believe their information is secure, their privacy is protected, and the system behaves transparently.

The findings therefore suggest that the success of FinGrator Pulse depends not only on technological innovation but also on its ability to establish credibility, transparency, and responsible financial intelligence.

3.0 PULSE SCORING FRAMEWORK

Pulse Score is one of the novel aspects of FinGrator Pulse, and its primary function is that of an index of financial health, helping users to gauge their current and projected financial standing. Unlike conventional credit scoring models that rely primarily on assessing the debtor's payment history, the Pulse Score model takes into account more holistic factors such as cash flow stability, expenditures, saving trends, and financial stability.

The suggested model of Pulse Score comprises the following major elements:

COMPONENT	WEIGHT
Cash Flow Stability	30%
Savings Behavior	20%
Spending Control	20%
Emergency Financial Buffer	15%

Debt Pressure	10%
Financial Goal Progress	5%

Cash Flow Stability receives the highest weighting because stable inflows and manageable outflows are the strongest indicators of financial sustainability. Savings Behavior and Spending Control are also heavily weighted because they reflect users' ability to maintain financial discipline and avoid excessive spending patterns.

The scoring system adopts a traffic-light model:

- Green indicates strong financial health and sustainable cash flow patterns.
- Yellow indicates moderate financial risk and possible future financial pressure.
- Red indicates severe financial vulnerability and high probability of financial stress.

This model aligns with recent behavioral finance research suggesting that individuals respond better to simplified financial signals and visual indicators than to highly technical financial information (Lusardi et al., 2021). Simplifying complex financial data into understandable indicators improves user engagement and financial awareness.

The survey findings also revealed that explainability is critical for trust. Users are unlikely to trust financial scores that appear arbitrary or unclear. Therefore, Pulse should provide explanations showing which financial behaviors influenced a user's score and what actions can improve financial health.

This recommendation is supported by recent Explainable Artificial Intelligence (XAI) research, which emphasizes transparency, interpretability, and accountability in AI-driven decision

systems (Ali et al., 2023). Users are more likely to trust predictive systems when recommendations can be explained clearly.

Another important finding relates to prediction accuracy. Seventy-four percent of respondents answered “Maybe” when asked whether they would continue using the application if financial predictions were occasionally inaccurate. This indicates that users may tolerate occasional forecasting errors provided the platform remains useful, transparent, and continuously improves.

This insight reflects recent findings in digital decision-making research which show that users are often willing to accept systems that provide satisfactory and useful outcomes even when predictions are not perfectly accurate (Dwivedi et al., 2021).

As a result, Pulse should position its predictive analytics as financial guidance tools rather than guaranteed forecasts. This realistic positioning will improve credibility and reduce unrealistic user expectations.

4.0 DATA USAGE MODEL AND PRIVACY ANALYSIS

The survey findings clearly demonstrate that trust in financial data management is central to user adoption. When respondents were asked what a financial application should never do with their data, the overwhelming majority emphasized concerns regarding unauthorized sharing, selling of personal information, data leaks, and misuse of financial records.

Common responses included:

- “Share it with third parties”

- “Sell it”
- “Leak confidential financial records”
- “Use my data for commercial purposes”
- “Track my activity secretly”

These responses indicate that users are highly sensitive to how their financial information is collected, stored, and shared. Recent privacy and fintech studies indicate that users weigh the benefits of personalized financial services against perceived privacy and cybersecurity risks before deciding whether to disclose sensitive financial information (Mhlanga, 2020).

To address these concerns, Pulse should adopt a consent-driven data usage framework built on five principles:

4.1 User Consent and Permission Control

Users should decide which financial data sources they wish to connect, including mobile money wallets, bank accounts, SMS alerts, or uploaded statements. Permissions should be adjustable and revocable at any time.

4.2 Data Minimization

Pulse should collect only information necessary for improving financial insights and forecasting accuracy. Excessive data collection may increase user distrust and cybersecurity exposure.

4.3 Encryption and Secure Infrastructure

Strong encryption, secure APIs, biometric authentication, and multi-factor authentication should be implemented to reduce risks of unauthorized access and cyber fraud.

This recommendation aligns with cybersecurity best practices proposed by the National Institute of Standards and Technology (NIST, 2020), which emphasizes layered security mechanisms for digital financial systems.

4.4 Transparency and Explainability

The platform should clearly explain:

- how user data is used,
- why recommendations are generated,
- and how predictions are calculated.

Transparency reduces suspicion and strengthens trust in AI-driven systems.

4.5. Ethical Data Governance

Pulse should never sell personal financial information to advertisers or third parties without explicit user consent. Ethical AI usage is critical because manipulative financial recommendations may damage user welfare and platform credibility.

The survey findings also revealed that institutional legitimacy significantly influences trust. Many respondents indicated that partnerships with banks, regulators, and recognized financial institutions would increase confidence in the platform. Recent digital finance studies similarly show that users rely heavily on institutional legitimacy, regulatory assurance, and compliance structures when evaluating fintech platforms (Anouze & Alamro, 2020).

5.0 SURVEY INSIGHTS AND USER BEHAVIOUR

ANALYSIS

The survey generated several important insights regarding user attitudes toward digital finance, AI-powered financial tools, privacy, and trust.

First, the findings indicate strong openness toward intelligent financial assistance. Fifty-eight percent of respondents stated they would use an AI-powered financial assistant, while 30% responded “Maybe.” This suggests growing acceptance of AI-supported financial guidance, particularly among digitally active younger users.

However, users do not appear ready for fully autonomous financial systems. When respondents were asked which type of financial advice they would trust most, 74% preferred a combination of AI and human advice, while very few preferred AI-only systems.

This finding supports recent human-centered AI studies which argue that effective technological systems should balance automation with human oversight and accountability (Shneiderman, 2020). Users value technological efficiency but still seek accountability, reassurance, and human judgment in financial decision-making.

The survey also highlighted the importance of personalization. Many respondents identified personalized recommendations, predictive analytics, spending analysis, and real-time financial updates as the most valuable features of a financial intelligence platform.

Recent digital consumer behavior studies similarly suggest that users are more likely to adopt technologies that satisfy their personal financial needs and improve convenience, personalization, and financial control (Nguyen et al., 2023). Users are therefore more likely to

engage with Pulse if the platform provides customized financial insights relevant to their spending habits and financial goals.

Another important insight relates to financial application adoption. Seventy-two percent of respondents indicated they had never previously used a budgeting or financial management application. This demonstrates that there remains a large untapped market for intuitive financial intelligence systems.

However, this also suggests that onboarding simplicity and trust-building mechanisms will be critical. Recent fintech adoption research shows that adoption of new digital financial technologies depends heavily on perceived usefulness, simplicity, compatibility with user lifestyles, and trust in the platform (Al Nawayseh, 2020).

Security emerged as the strongest recurring theme throughout the survey. Respondents consistently identified data privacy, fraud prevention, security systems, and transparency as their primary concerns.

This confirms that trust is not a secondary feature of fintech systems but rather a foundational requirement for user adoption and retention.

Finally, the survey findings reflect the growing dominance of mobile-first financial ecosystems in Ghana. The widespread use of mobile money suggests that Pulse should prioritize integration with mobile money services rather than focusing exclusively on traditional banking infrastructure.

6.0 CONCLUSION

FinGrator Pulse represents a shift from traditional budgeting applications toward intelligent financial decision-support systems. The survey findings demonstrate that users are increasingly open to AI-powered financial guidance and predictive analytics, particularly within mobile-first financial ecosystems such as Ghana.

However, the findings also confirm that trust, transparency, and security are essential conditions for adoption. Users are willing to engage with intelligent financial systems only when they believe their information is protected and used responsibly.

The proposed Pulse Score framework, consent-driven data usage model, and trust-centered strategy position Pulse as a practical fintech solution capable of improving financial visibility, financial planning, and long-term financial decision-making.

Rather than functioning simply as a transaction tracker, Pulse has the potential to become a responsible financial intelligence ecosystem that helps users move from financial uncertainty toward financial clarity and control.

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